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Activity - 2

Stack queue and Linked List Question

① stack and Queue

stack operations:

- push (54) → [54]
- push (52) → [54, 52]
- pop() → removes 52
- push (55) → [54, 55]
- push (62) → [54, 55, 62]
- s = pop() → 62

s = 62

Queue operations:

- enqueue (21) → [21]
- enqueue (24) → [21, 24]
- dequeue() → remove 21
- enqueue (28) → [24, 28]
- enqueue (32) → [24, 28, 32]
- q = dequeue () → 24

q = 24

s+q = 62+24 = $\boxed{86}$

② Postfix Expression

10 5 + 60 6 / * 8 -

10 5 + = 15

60 6 / = 10

15 x 10 = 150

150 - 8 = 142

Ans $\boxed{42}$

③

- The queue is reversed by removing element from the front and inserting them back using enqueue operations.
- The maximum number of enqueue operations required to reverse the given queue is 4. Answer : 4.

④ * The reversed operation can be simulated using only push and pop operation.
* The stack element are poped and pushed back in reverse order.

⑤ Expression : 8 2 3 ^ / 2 3 * + 5 1 * -

* 2 ^ 3 = 8

stack: 8 8

* 8 / 8 = 1

stack = 1

* 2 * 3 = 6

stack = 1 6

* 5 * 1 = 5

stack : 7 5

* 7 - 5 : 2

stack = 2

Final answer = 2

① Infix : a+b*c - d^e^f

b*c = bc*

a + bc* = abc*+

e^f = ef^

d^ef^ = def^^

abc*+ - def^^ = abc*+def^^-

postfix: abc*+def^^-

⑨ Operation : push (10), push (20), pop, push (10), push (20), pop, pop, push (20) pop

push (10) → [10]

push (20) → [10, 20]

pop → 20

push (10) → [10, 10]

push (20) → [10, 10, 20]

pop → 20

pop → 10

pop → 10

push (20) → 20

pop → 20

Values poped in order : 20, 20, 10, 10, 20

Values pushed in order : 10, 20, 10, 20, 20

Good

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